

PROJECT REPORT

EduStream

A Course Enrollment & Learning Management System

Capstone Project — Task 5

Full Stack Web Development Internship (PHP & MySQL)

Submitted to: ApexPlanet Software Pvt. Ltd.

Prepared by: G. Praveen

Live Application: <http://edustream.kesug.com>

GitHub Repository: <https://github.com/G-Praveen88/task5-elearning>

September 2026

1. Executive Summary

EduStream is a full-stack e-learning portal built in PHP and MySQL as the capstone project (Task 5) of the ApexPlanet Software Full Stack Web Development internship. The platform allows students to browse a course catalog, enroll in courses, and track lesson-by-lesson progress, while administrators and instructors manage the course catalog, curriculum, enrolled students, and platform analytics through a dedicated admin console.

The project was designed, developed, and deployed independently across all required phases: requirements planning, database design, secure authentication with email OTP verification, core student-facing modules, an administrative management suite, and live deployment to public hosting.

Key Highlights

- Fully functional Student and Admin/Instructor roles with role-based routing
- Email OTP-based account verification (simulated inbox for grading, clearly documented)
- AJAX-powered course search, filtering, and lesson completion — no full page reloads
- Five-page Admin Console including a Chart.js analytics dashboard (line, bar, doughnut charts)
- Production-grade security: prepared statements, CSRF tokens, bcrypt hashing, hardened file uploads
- Live-deployed on free hosting (InfinityFree) and version-controlled via a public GitHub repository

2. Project Overview & Objectives

The objective of Task 5 was to independently design, build, and deploy a professional-grade capstone project demonstrating readiness for industry full-stack development work. EduStream was selected as a unique project idea: an e-learning portal covering course discovery, enrollment, progress-tracked lesson delivery, and a full administrative back office — deliberately exceeding the minimum scope to also include an analytics dashboard.

Objectives

- Design a normalized relational schema supporting users, courses, enrollments, and lessons
- Implement secure registration and login with OTP-based email verification
- Build a responsive, mobile-first student experience for browsing and learning
- Build a complete administrative suite for content and student management
- Apply industry-standard security practices throughout the codebase
- Deploy the finished application to live, publicly accessible hosting

3. User Roles & Use Cases

3.1 Student

- Register for an account and verify email via a 6-digit OTP
- Browse the course catalog with real-time search and category filtering
- View course details, syllabus, and instructor information
- Enroll in a course and begin the first lesson immediately
- Watch lesson content and mark lessons complete, with live progress tracking

- View a personal dashboard summarizing enrolled courses and completion status

3.2 Instructor / Administrator

- Sign in to a dedicated admin console separate from the student experience
- View platform-wide KPIs: total students, courses, enrollments, weekly growth, revenue
- Create, edit, and delete courses, including validated thumbnail image uploads
- Manage a course's lesson sequence, with automatic ordering and progress recalculation
- Search and filter enrolled students by name, email, or course, with progress breakdowns
- View visual analytics: enrollment trends, popular courses, and category distribution

4. Technology Stack

Layer	Technology
Backend	PHP 8, procedural with mysqli prepared statements
Database	MySQL (InfinityFree-hosted in production)
Frontend	Bootstrap 5.3, vanilla JavaScript, AJAX (Fetch API)
Charts	Chart.js (line, bar, doughnut visualizations)
Security	bcrypt (cost 12), CSRF tokens, session regeneration, finfo MIME validation
Hosting	InfinityFree (free PHP + MySQL hosting)
Version Control	Git & GitHub (public repository)

5. Database Schema

The database consists of four core tables with foreign key relationships enforcing referential integrity, plus indexes on frequently filtered columns (category, user_id, course_id).

5.1 Entity Overview

Table	Purpose	Key Columns
users	Stores student & instructor/admin accounts	id, name, email, password_hash, role, otp_code, otp_verified, created_at
courses	Course catalog entries	id, title, description, category, price, instructor_id (FK), thumbnail, created_at
enrollments	Links students to courses with progress	id, user_id (FK), course_id (FK), enrolled_at, progress_percent
lessons	Curriculum content within a course	id, course_id (FK), title, content, video_url, order_index

5.2 Relationships

- A user with role = instructor/admin can author many courses (one-to-many: users → courses)

- A course can have many enrollments; a user can have many enrollments (many-to-many via enrollments)
- A course can have many lessons, ordered by `order_index` (one-to-many: courses → lessons)
- Deleting a lesson triggers automatic recalculation of `progress_percent` for every enrolled student in that course

6. Features Implemented

6.1 Authentication & Onboarding

- Registration with bcrypt password hashing (cost factor 12)
- 6-digit numeric OTP generated via `random_int()`, with a 10-minute expiry and single-use invalidation
- On-screen simulated email inbox preview, since no SMTP server is configured (clearly commented in code)
- Session fixation defense via `session_regenerate_id(true)` on successful login
- Role-based redirect after login: Student → dashboard, Admin/Instructor → admin console

6.2 Student-Facing Modules

- Homepage with live platform statistics and featured courses
- Course catalog with debounced AJAX search and category filter pills, no page reloads
- Course detail page with syllabus, instructor info, and dynamic enrollment call-to-action
- CSRF-protected enrollment processor redirecting directly into the first lesson
- Lesson player with responsive video embed, lecture notes, and AJAX-based completion toggle
- Student dashboard with KPI widgets and live-updating Bootstrap progress bars

6.3 Admin & Instructor Console

- Executive dashboard: 5 real-time KPI tiles plus a live recent-enrollment feed
- Full course CRUD with hardened thumbnail upload validation (size limit, extension whitelist, finfo MIME check, randomized filenames)
- Lesson sequence manager with add/edit/delete and automatic order-index handling
- Student progress roster with search and course filtering
- Analytics dashboard: 14-day enrollment trendline, popular-course bar chart, category-share doughnut chart

7. Security Implementation

Security was treated as a first-class requirement throughout development, not an afterthought:

- 100% parameterized queries via mysqli prepared statements — no raw string interpolation into SQL anywhere in the codebase
- CSRF tokens (32-byte, `hash_equals-verified`) on every POST form
- Passwords hashed with bcrypt at cost factor 12
- Output escaping via a centralized `e()` helper wrapping `htmlspecialchars()` to prevent XSS
- File uploads validated on three layers: file size ($\leq 2\text{MB}$), extension whitelist, and binary MIME-type inspection via PHP's `finfo` — with randomized, sanitized filenames on save
- OTP codes are numeric, time-boxed to 10 minutes, and single-use

- Session ID regenerated on every successful login to prevent session fixation attacks

8. Application Screenshots

The screenshots below were captured from the live, deployed application at <http://edustream.kesug.com>.

8.1 Admin Executive Dashboard

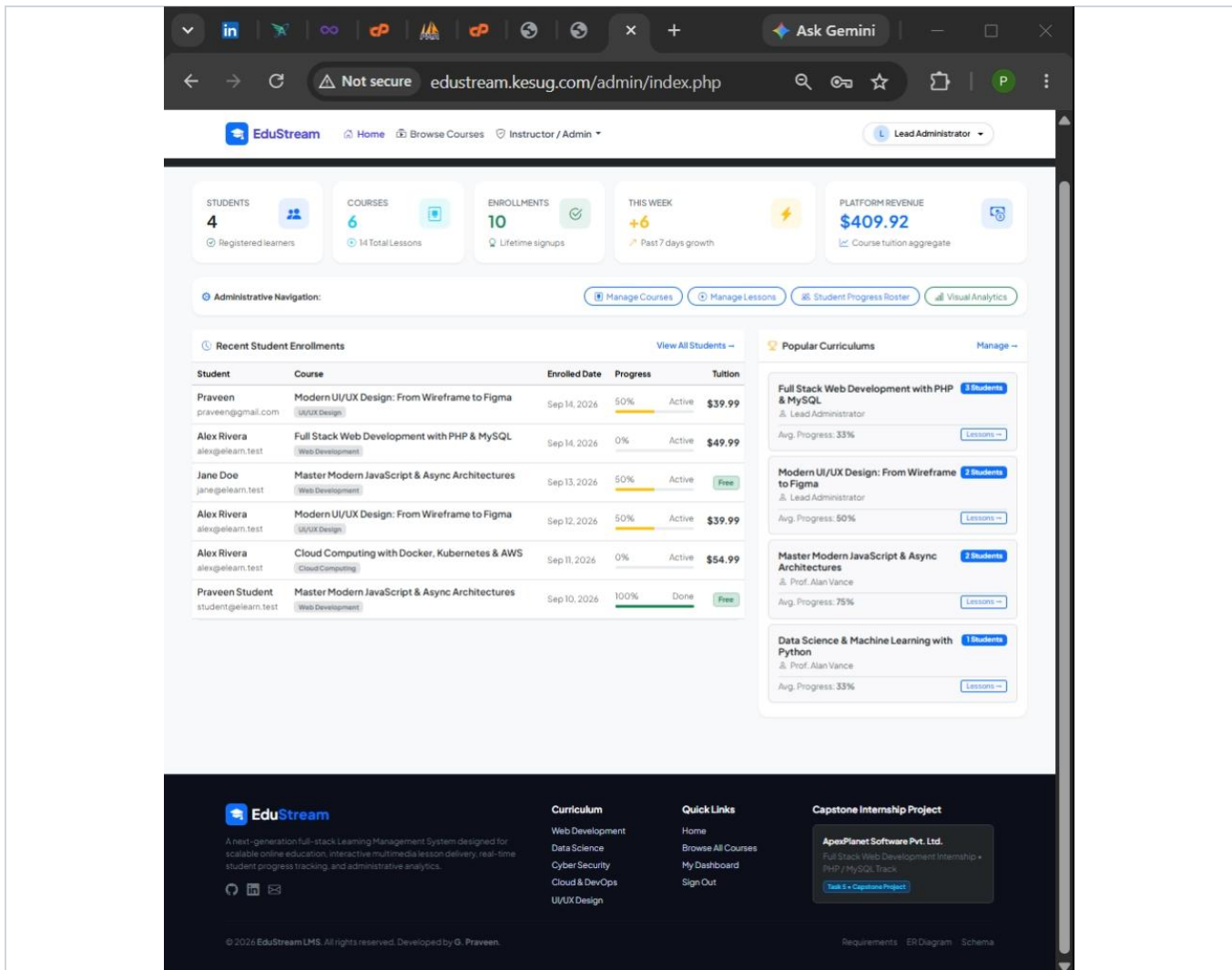


Figure 1 — Admin dashboard showing live KPIs, recent enrollments, and popular curricula

8.2 Homepage

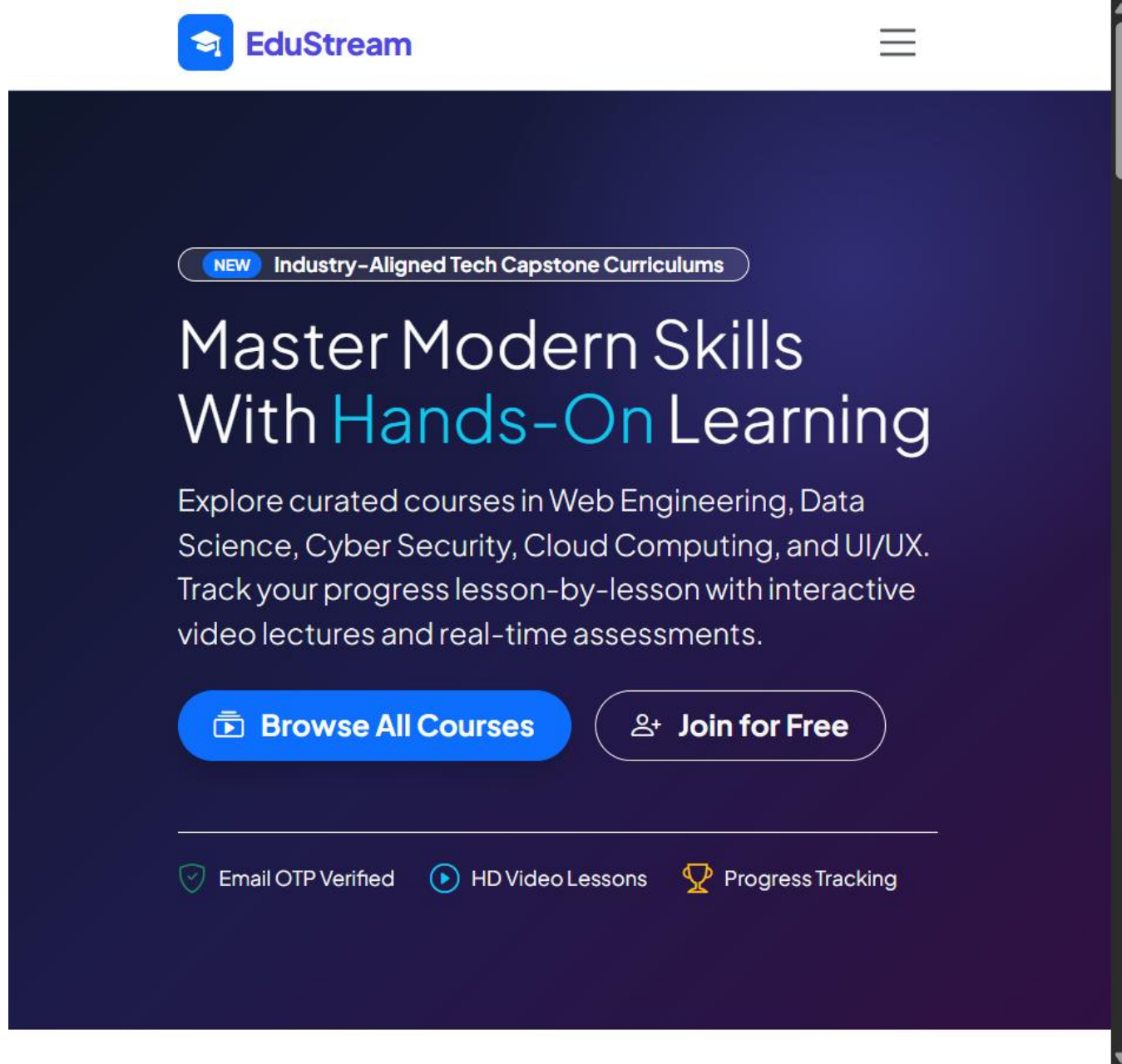


Figure 2 — Homepage with platform statistics and featured courses

8.3 Registration & OTP Verification

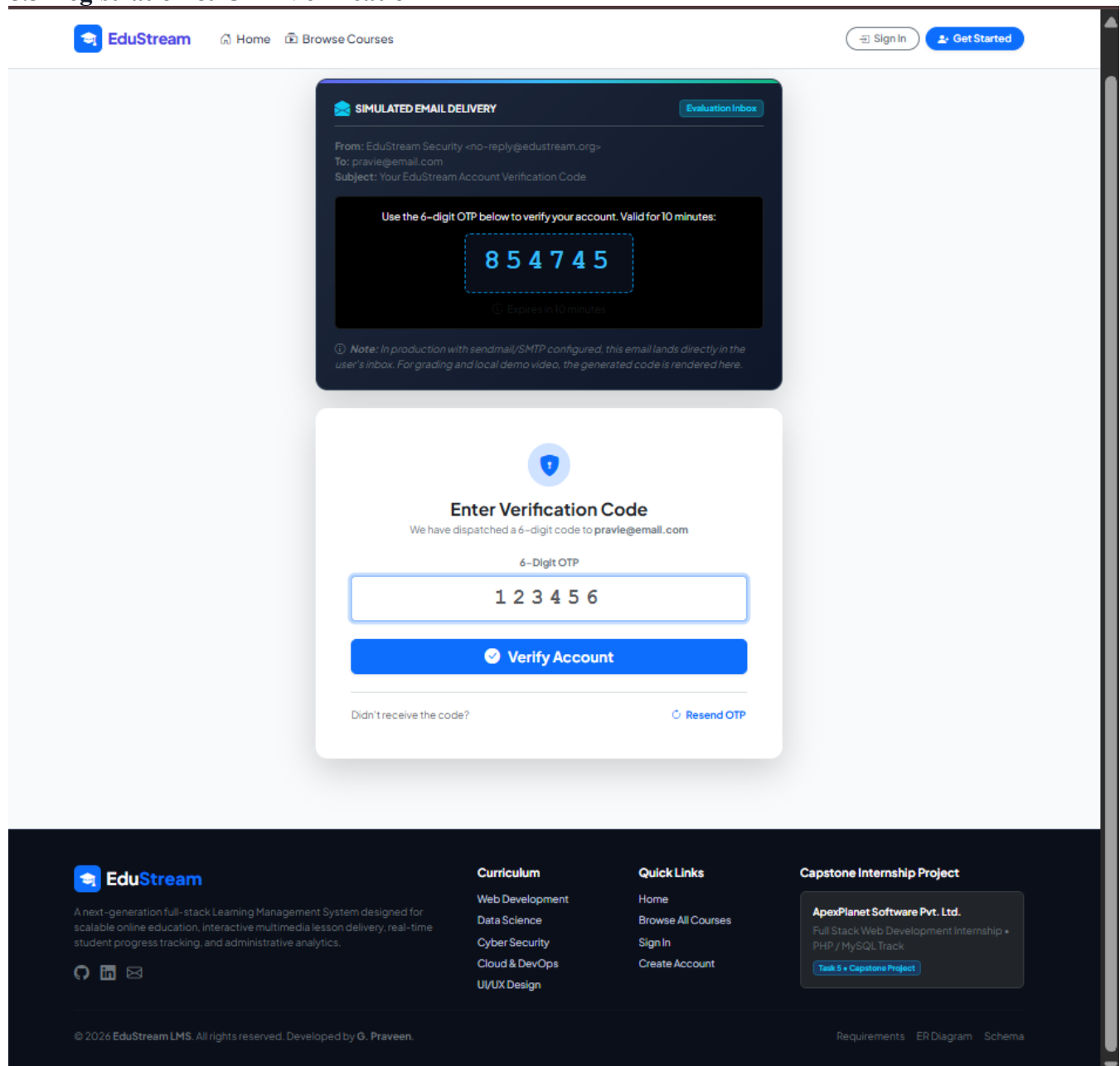


Figure 3 — Registration form and simulated OTP email delivery

8.4 Student Dashboard

 **EduStream**

[Home](#) [Browse Courses](#) [My Learning](#)

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Praveen 

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Welcome back, Praveen!
Track your courses, resume lessons, and achieve your learning targets.

 [Browse More Courses](#)

ENROLLED COURSES

1



IN PROGRESS

1



COMPLETED

0



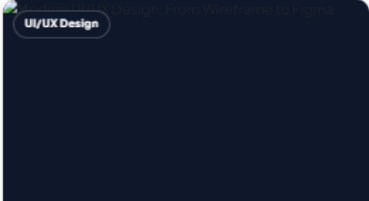
COMPLETED LESSONS

1



My Enrolled Courses
Continue your curriculum right where you left off

1 Active Enrolled



Enrolled Sep 14, 2026

In Progress

Modern UI/UX Design: From Wireframe to Figma

Instructor: Lead Administrator

Curriculum Progress

50%

1 of 2 Lessons Done

[Continue Learning](#)

 **EduStream**
A next-generation full-stack Learning Management System designed for scalable online education, interactive multimedia lesson delivery, real-time student progress tracking, and administrative analytics.


Curriculum

Web Development

Data Science

Cyber Security

Cloud & DevOps

UI/UX Design

Quick Links

[Home](#)

[Browse All Courses](#)

[My Dashboard](#)

[Sign Out](#)

Capstone Internship Project

ApexPlanet Software Pvt. Ltd.
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[Task 5 • Capstone Project](#)

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[Requirements](#) [ER Diagram](#) [Schema](#)

Figure 4 — Student dashboard with enrolled course progress

8.5 Course Catalog & Lesson Player

 **EduStream**

[Home](#) [Browse Courses](#) [My Learning](#)

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Course Catalog

Browse interactive courses, comprehensive syllabuses, and video curriculums

All Categories

Cloud Computing

Cyber Security

Data Science

UI/UX Design

Web Development

Web Development

2 Lessons2 Students

Master Modern JavaScript & Async Architectures
Deep dive into ES6+, closures, prototype chains, asynchronous event loops, Promises, Fetch API, and modular....

Prof. Alan Vance

Free

View Syllabus & Enroll

UI/UX Design

2 Lessons2 Students

Modern UI/UX Design: From Wireframe to Figma
Master human-centered digital design, interactive prototyping, color theory, responsive grid layouts, and d...

Lead Administ...

\$39.99

View Syllabus & Enroll

Cloud Computing

2 Lessons1 Students

Cloud Computing with Docker, Kubernetes & AWS
Deploy, orchestrate, and scale enterprise container workloads. Covers AWS EC2, S3, Dockerfile optimization,...

Prof. Alan Vance

\$54.99

View Syllabus & Enroll

Cyber Security

2 Lessons1 Students

Cyber Security & Ethical Hacking Essentials
Learn network defensive architectures, vulnerability scanning, OWASP Top 10 mitigation, web penetration tes...

Lead Administ...

\$64.99

View Syllabus & Enroll

Data Science

3 Lessons1 Students

Data Science & Machine Learning with Python
A comprehensive journey through Python data analysis, NumPy, Pandas, statistical modeling, Scikit-learn pre...

Prof. Alan Vance

\$59.99

View Syllabus & Enroll

Web Development

3 Lessons3 Students

Full Stack Web Development with PHP & MySQL
Master modern web engineering from relational database design and secure PDO/MySQLi querying to responsive ...

Lead Administ...

\$49.99

View Syllabus & Enroll

Figure 5 — Course browsing with AJAX search and the lesson video player

8.6 Analytics Dashboard

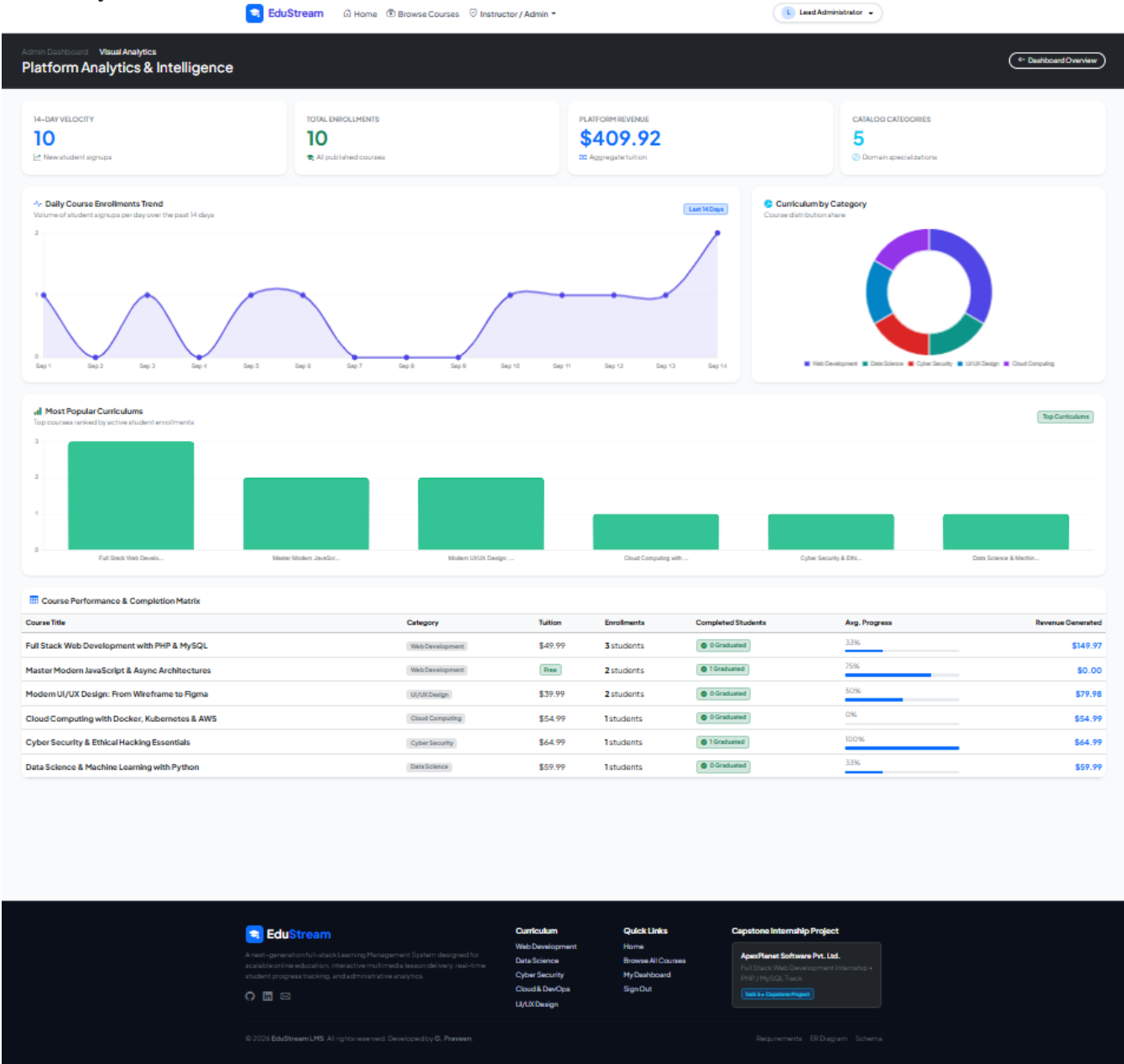


Figure 6 — Chart.js analytics: enrollment trend, popular courses, category share

9. Deployment

Hosting Provider: InfinityFree (free PHP + MySQL hosting)

Live URL: <http://edustream.kesug.com>

Source Repository: <https://github.com/G-Praveen88/task5-elearning>

The application was deployed by provisioning a MySQL database on InfinityFree, importing schema.sql via phpMyAdmin, updating database credentials in config/db_connect.php for the live environment, and uploading the project files into the htdocs directory. The deployed site was manually verified end-to-end: registration, OTP verification, login, course browsing, enrollment, lesson completion, and every admin console page were tested live and confirmed working.

10. Challenges & Learnings

- Path resolution bug: the `base_url()` helper correctly detected a `/task5-elearning` subfolder prefix in the local XAMPP environment, but on live hosting (where the app is served from the domain root) it fell back to using the current page's folder as the base path, causing doubled URLs such as `/auth/auth/register.php`. This was resolved by defaulting to an empty base path when no subfolder prefix is detected.
- Learned the practical difference between local development paths and production root-relative paths, and the importance of testing every navigation link after deployment, not just the homepage.
- Gained hands-on experience with the full free-hosting deployment workflow: database provisioning, schema import via phpMyAdmin, and file upload via a web-based file manager.

11. Conclusion & Future Enhancements

EduStream successfully delivers a complete, secure, and functional e-learning platform covering the full student journey from registration to course completion, alongside a full administrative suite for content and analytics management. All core and security requirements for Task 5 were met and verified on a live deployment.

Future Enhancements

- Integrate real SMTP email delivery in place of the simulated on-screen OTP
- Add payment gateway integration for paid courses
- Support native video hosting/upload instead of embedded YouTube links
- Add automated CI/CD deployment from GitHub for faster iteration